Kori Vernon

Project Portfolio

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Personal trading and engineering projects built outside my role at Morgan Stanley Electronic Trading, operated within firm personal trading restrictions. View-only links to live PnL and performance dashboards are available on request.

PomTrader

*Python · E*TRADE API*

A personal trading platform that manages my brokerage accounts end to end: systematic strategy execution, hedge suggestions and tax-efficiency logic in one system.

- Runs a live systematic strategy on bull and bear levered ETFs using MACD/RSI momentum signals with custom parameters and entry/exit rules, with automated execution through the E*TRADE API; live Sharpe ratio of 4.3 since inception.
- Built a backtesting, paper-trading and forward-testing framework to develop and tune new strategies before promoting them to live capital.
- Suggests portfolio hedges (ETF, options and beta hedges) against current positions.
- Applies tax-efficiency strategies to position management and trade decisions.

Sports Prediction Market Maker

Python · Polymarket

An automated market maker that quotes two-sided markets on sports event contracts.

- Quotes across all sports markets on Polymarket. Initially ran on both Kalshi and Polymarket, then consolidated onto Polymarket after the data showed it was more profitable and better suited to making markets.
- Derives fair value from a proprietary pricing model and quotes around it, managing spread and exposure across events.
- Applies market-making concepts I first saw supporting an automated equities market-making desk and have since developed through self-study and building this system: spread capture, adverse selection and inventory risk.

EMon (Equities Monitor)

*Python · Flask · SQL · E*TRADE API*

A research and screening platform that scans roughly 6,900 U.S.-listed stocks every 15 minutes, flags unusual behavior, turns flags into paper-tracked stock and options positions, and reports through a web app and daily emails.

Ivormo: volatility outlier mean-reversion strategy

- Flags moves that are extreme relative to each stock's own volatility, measured in typical-day sigma units.
- Killed the original options book on the evidence (Sharpe ≈ -2.5; buying options lost after costs at every tenor and strike tested) and rebuilt it from 9,593 historical flags and 3,384 priced on real option chains.
- Stock leg: walk-forward ridge regression on move size, gap share, run-up, distance from 52-week high and SPY's move; out of sample (2021–2026) +5.0% per trade, 71% hit rate, $t = 8.3$, every year positive; daily-book Sharpe of 1.2 with volatility-scaled sizing.
- Options leg: sells defined-risk credit spreads only when IV/RV ≥ 1.28, with results recorded at both mid and natural prices so bid/ask cost is measured, not assumed.

Earnings equity options screener

- Compares the options-implied earnings move (at-the-money straddle, with non-earnings variance stripped out) against a model-expected move fit on 18,600+ historical reactions, and labels each name cheap or rich.
- Prices defined-risk iron condor ideas by simulating each stock's historical earnings-move distribution net of bid/ask costs; ideas are paper-tracked and scored after the print.
- Finding: move size is predictable (rank correlation ≈ 0.42) but already priced; implied moves averaged 1.06× realized across 2,188 events.